

# Ahatsham Hayat

Postdoctoral Research Fellow, Department of Anesthesia – Research  
Harvard Medical School, Harvard University, USA

✉ Preferred Email: [afnu@bwh.harvard.edu](mailto:afnu@bwh.harvard.edu)  [ahatsham.github.io](https://github.com/ahatsham)  Google Scholar  
📍 Local Address: 2 Oakland Street Apt. 2, Brighton, MA 02135, USA

## Research Interests and Lab Work

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My research focuses on developing generalizable, reliable, and clinically meaningful multimodal AI systems for complex longitudinal data. My work spans large language models, vision-language models, NLP, time-series forecasting, multimodal learning, computer vision, and behavioral modeling, with emphasis on models that remain robust across populations, cohorts, and distribution shifts.

In Dr. Vesela Kovacheva's lab, I will develop predictive models to improve maternal patient safety and prevent adverse pregnancy outcomes. This work will use multidimensional clinical data, including ECG/EEG waveform signals, structured EHR data, genetic information, imaging, and clinical text, to build NLP-based, time-series, and multimodal models for predicting hypertensive crises, hemodynamic instability, hemorrhage, and ICU admission. The goal is to create accurate, interpretable, and clinically actionable AI systems for integration into clinical practice.

I have authored **35 peer-reviewed publications** and have experience working on **NSF, NIH**, and collaborative research grants.

## Education

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### Ph.D. in Computer Engineering

Dates Attended: 2023–2026

University of Nebraska–Lincoln, Lincoln, Nebraska, USA

Completed: 2026 GPA: 4.0/4.0

Thesis: *A Multimodal Approach for Building Generalizable and Reliable Models of Longitudinal Experiential Data*

### M.S. in Telecom Engineering

Dates Attended: 2025–2025

University of Nebraska–Lincoln, Lincoln, Nebraska, USA

Completed: 2025 GPA: 4.0/4.0

### M.Tech in Computer Science

Dates Attended: 2014–2016

Visvesvaraya National Institute of Technology (VNIT), India

Completed: 2016 GPA: 7.5/10

Thesis: *Stream Clustering Algorithm for Detecting Temporal Contexts*

### B.Tech in Computer Science

Dates Attended: 2010–2014

Uttarakhand Technical University, India

Completed: 2014    Grade: 70%  
Thesis: *Fake News Detection using Binary Classification*

## Work Experience

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### Postdoctoral Research Fellow

June 2026 – Present

Brigham and Women's Hospital (BWH), Department of Anesthesia – Research, Harvard Medical School, USA

- Develop predictive models to enhance maternal patient safety and prevent adverse pregnancy outcomes.
- Analyze multidimensional clinical datasets, including ECG/EEG waveform signals, imaging, genetic data, structured EHR data, and clinical text.
- Build NLP-based, time-series, and multimodal learning models for risk prediction of hypertensive crises, hemodynamic instability, hemorrhage, and ICU admission.
- Contribute to clinically actionable AI systems designed for integration into real-world clinical practice.

### Graduate Research Assistant

Jan 2023 – May 2023

University of Nebraska–Lincoln, USA

- Lead researcher on NSF-funded projects in AI-enabled educational feedback and behavioral modeling.
- Developed multimodal architectures integrating LLMs, VLMs, and time-series models.
- Created narrative-based LLM pipelines for cross-cohort generalization.
- Designed automated LLM educational-feedback and behavioral reasoning systems.

### Research Assistant

Nov 2021 – Jan 2023

Madeira Interactive Technologies Institute (MITI), Portugal

- Built computer-vision models and sensing systems for agricultural applications (BASE Project).
- Developed banana yield prediction and automated harvesting support systems.

### Assistant Professor (Research)

Nov 2017 – Aug 2021

University of Petroleum and Energy Studies (UPES), India

- Taught core and advanced courses in machine learning, deep learning, AI, and algorithms. Complete course list available in the **Teaching Experience** section.
- Supervised undergraduate and postgraduate thesis projects.
- Published research in applied machine learning and computer vision.

### Assistant Professor (Research)

July 2016 – Nov 2017

Parul University, India

- Taught foundational courses in programming, data structures, algorithms, and introductory AI. Full teaching record listed in the **Teaching Experience** section.
- Supervised undergraduate research and contributed to curriculum development.

- Mentored students in applied ML and introductory computer vision.

## Teaching Experience

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| <b>Fall 2016–2017</b>   | Operating Systems; Introduction to Artificial Intelligence (Parul University, India)                                      |
| <b>Spring 2016–2017</b> | Introduction to Object-Oriented Programming (Parul University, India)                                                     |
| <b>Fall 2017–2018</b>   | Introduction to Machine Learning (University of Petroleum and Energy Studies, India)                                      |
| <b>Spring 2017–2018</b> | Data Structures & Algorithms; Introduction to Operating Systems (University of Petroleum and Energy Studies, India)       |
| <b>Fall 2018–2019</b>   | Introduction to Machine Learning; Automata Theory (University of Petroleum and Energy Studies, India)                     |
| <b>Spring 2018–2019</b> | Data Structures & Algorithms (University of Petroleum and Energy Studies, India)                                          |
| <b>Fall 2019–2020</b>   | Introduction to Machine Learning; Introduction to Deep Learning (University of Petroleum and Energy Studies, India)       |
| <b>Spring 2019–2020</b> | Introduction to Artificial Intelligence; Data Structures & Algorithms (University of Petroleum and Energy Studies, India) |
| <b>Spring 2020–2021</b> | Introduction to Machine Learning; Introduction to Deep Learning (University of Petroleum and Energy Studies, India)       |
| <b>Fall 2020–2021</b>   | Introduction to Artificial Intelligence; Data Structures & Algorithms (University of Petroleum and Energy Studies, India) |

## Student Mentorship

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- **Hunter Tridle**  
B.S. in Electrical and Computer Engineering, University of Nebraska–Lincoln  
Mentored: Summer 2025 – Fall 2025
- **Helen Martinez**  
B.S. in Mathematics, University of Nebraska–Lincoln  
Mentored: Fall 2024 – Spring 2025
- **Sharif Akil**  
B.S. in Electrical and Computer Engineering, University of Nebraska–Lincoln  
Mentored: Fall 2023 – Spring 2024
- **Nathan Roberts**  
B.S. in Electrical and Computer Engineering, University of Nebraska–Lincoln  
Mentored: Spring 2023

## Books Edited

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Industry 5.0: Emerging Paradigms for Business, Technology, and Society

CRC Press, 2025

Editors: D. Sharma, M. Gupta, R. K. Kovid, T. P. Singh, P. Kumar, **A. Hayat**

## Publications

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### 2026 (Submitted)

1. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *Cross-Distribution Generalization in Longitudinal Behavioral Data Through Frozen Coherence Constraints*, Submitted to *Conference on Neural Information Processing Systems (NeurIPS 2026)*.
2. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *Cross-Cohort Generalization in Behavioral Data Requires Semantic Anchoring*, Submitted to *Conference on Neural Information Processing Systems Position Paper Track (NeurIPS 2026)*.

### 2026

3. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *Complementing Text Serialization with Visual Encoding of Structured Temporal Data for Cross-Distribution Generalization*, *Conference on Empirical Methods in Natural Language Processing (EMNLP Findings 2026)*.
4. H. A. Diefes-Dux, **Hayat Ahatsham**, M. R. Hasan, *WIP: AI Tool for Theory-Aligned Grading and Feedback of Student Reflections, Preliminary Findings for Monitoring*, *IEEE*, October 2026.
5. **Ahatsham Hayat**, Neeta Kantamneni, Bilal Khan, Mohammad Rashedul Hasan, *Human-Guided Generative AI for Proactive STEM Student Support*, *The 2026 ASEE Annual Conference*.
6. **Ahatsham Hayat**, Hunter Tridle, Mohammad Rashedul Hasan, *From Numbers to Narratives: Efficient Language Model-Based Detection for Safety-Critical Minority Classes*, *EACL Findings 2026*.

### 2025

7. **Ahatsham Hayat**, Bilal Khan, Mohammad Rashedul Hasan, *ConText-LE: Cross-Distribution Generalization for Longitudinal Experiential Data via Narrative-Based LLM Representations*, *Conference on Empirical Methods in Natural Language Processing (EMNLP 2025) Findings*.
8. **Ahatsham Hayat**, Helen Martinez, Bilal Khan, Mohammad Rashedul Hasan, *A Three-Tier LLM Framework for Forecasting Student Engagement from Qualitative Longitudinal Data*, *The 29th Conference on Computational Natural Language Learning (CoNLL 2025)*.
9. **Ahatsham Hayat**, Helen Martinez, Bilal Khan, Mohammad Rashedul Hasan, *Harnessing Language Models to Predict and Enhance STEM Engagement Using Non-Cognitive Experiential Data*, *The 2025 ASEE Annual Conference*.
10. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *A Context-Aware Approach for Enhancing Data Imputation with Pre-trained Language Models*, *The 31st International Conference on Computational Linguistics (COLING 2025)*.

11. A. Haleem, M. Javaid, **Ahatsham Hayat**, R. Suman, S. Rab, R. P. Singh, *Cloud computing applications for Industry 4.0: A literature-based study*, *Journal of Industrial Integration and Management*, World Scientific Publishing Company.
12. P. Baglat, **Ahatsham Hayat**, S. S. Mostafa, F. Mendonça, F. Morgado-Dias, *Comparative analysis and evaluation of YOLO generations for banana bunch detection*, *Smart Agricultural Technology*, Volume 12, 2025, 101100, ISSN 2772-3755.

## 2024

13. **Ahatsham Hayat**, Bilal Khan, Mohammad Rashedul Hasan, *Leveraging Language Models for Analyzing Experiential Time-Series Data in Education*, *International Conference on Machine Learning and Applications (ICMLA)*.
14. **Ahatsham Hayat**, Bilal Khan, Mohammad Hasan, *Enhancing Transfer Learning for Early Forecasting of Academic Performance by Contextualizing Language Models*, *The 19th NAACL Workshop on Innovative Use of NLP for Building Educational Applications (BEA)*.
15. **Ahatsham Hayat**, Sharif Wayne Akil, Helen Martinez, Bilal Khan, Mohammad Rashedul Hasan, *Enhancing Zero-Shot Learning of Large Language Models for Early Forecasting of STEM Performance*, *The 2024 ASEE Annual Conference*.
16. **Ahatsham Hayat**, F. Morgado-Dias, T. Choudhury, T. P. Singh, K. Kotecha, *FruitVision: A deep learning based automatic fruit grading system*, *Open Agriculture*, vol. 9, no. 1, 2024, pp. 20220276.
17. **Ahatsham Hayat**, P. Baglat, F. Mendonça, S. S. Mostafa, F. Morgado-Dias, *Machine learning system for commercial banana harvesting*, *Engineering Research Express* 6, 035202.

## 2023

18. **Ahatsham Hayat**, Mohammad Rashedul Hasan, *The Power of Personalization and Contextualization: Early Student Performance Forecasting with Language Models*, *The 2023 NeurIPS Workshop on Generative AI for Education (GAIED)*.
19. H. K. Sharma, H. F. Mahdi, T. Choudhury, **Ahatsham Hayat**, *Automated System Configuration Using DevOps Approach*, *Proceedings of Fourth International Conference on Computer and Communication Technologies*, Lecture Notes in Networks and Systems, vol 606, Springer, Singapore.
20. P. Baglat, **Ahatsham Hayat**, F. Mendonca, A. Gupta, S. S. Mostafa, F. Morgado-Dias, *Non-Destructive Banana Ripeness Detection Using Shallow and Deep Learning: A Systematic Review*, *Sensors*, 2023; 23(2):738.
21. I. V. Manoj, S. Y. Narendranath, P. M. Mashinini, H. Soni, S. Rab, S. Ahmad, **Ahatsham Hayat**, et al., *Artificial neural network-based prediction assessment of wire electric discharge machining parameters for smart manufacturing*, *Paladyn, Journal of Behavioral Robotics*, Vol. 14 (Issue 1), pp. 20220118.

22. **Ahatsham Hayat**, *Breast Cancer Detection System from Thermal Images using SWIN Transformer*, *Journal Press India*, June 15, 2023.
23. V. Shahare, N. Arora, **A. Hayat**, N. Mouje, *Machine Learning Algorithms for Big Data Analytics, Demystifying Big Data Analytics for Industries and Smart Societies*. 2023.
24. **A. Hayat**, V. Shahare, A. K. Sharma, N. Arora, *Introduction to Industry 4.0, Blockchain and Its Applications in Industry 4.0*. 2023.
25. **A. Hayat**, P. Baglat, F. Mendonça, S. S. Mostafa, F. Morgado-Dias, *Novel Comparative Study for the Detection of COVID-19 Using CT Scan and Chest X-Ray Images*, *International Journal of Environmental Research and Public Health* 20 (2), 1268. 2023.

## 2022

26. **Ahatsham Hayat**, F. Morgado-Dias, *Deep Learning-Based Automatic Safety Helmet Detection System for Construction Safety*, *Applied Sciences* 12, no. 16: 8268.
27. **Ahatsham Hayat**, F. Morgado-Dias, B. P. Bhuyan, R. Tomar, *Human Activity Recognition for Elderly People Using Machine and Deep Learning Approaches*, *Information* 13, no. 6: 275.
28. N. Silva, G. Machado, I. Viveiros, H. Garças, F. Silva, R. Sousa, R. Gouveia, **A. Hayat**, et al., *Agrotech–Sensing Use Cases*, 2022.

## 2020–2018

29. **Ahatsham Hayat**, S. Jain, V. Shahare, N. Arora, S. Rab, *Real Time Human Activity Recognition Using Smart Phone*, *Advanced Science, Engineering and Medicine*, Volume 12, Number 9, September 2020, pp. 1200–1203(4), American Scientific Publishers.
30. K. Singh, K. Kaushik, **A. Hayat**, V. Shahare, *Role and Impact of Wearables in IoT Healthcare*, *Proceedings of the Third International Conference on Computational Intelligence*. 2020.
31. V. Shahare, N. Arora, **A. Hayat**, N. Mouje, *Smart Home Automation Tool for Energy Conservation*, *International Journal of Recent Technology and Engineering* 8, 5312–5315. 2019.
32. **Ahatsham Hayat**, A. Singh, V. Shahare, N. Arora, *An Efficient System for Early Diagnosis of Breast Cancer using Support Vector Machine*, *International Journal of Engineering and Advanced Technology*. 2019.
33. **Ahatsham Hayat**, N. Arora, K. P. Singh, *An Approach towards Real Time Smart Vehicular System Using Internet of Things*, *International Journal of Research in Engineering, IT and Social Sciences*. 2018.
34. N. Arora, **A. Hayat**, K. Singh, *Achieving Security in Medical Scanned Images Using Extended Image Steganography*, 2018.

## Professional Service

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- **Organizer**, Madeira International Workshop in Machine Learning, Funchal, Portugal (July 18–22, 2022)
- **Reviewer**, American Society for Engineering Education (ASEE) (2024–2026)
- **Reviewer**, Frontiers in Education Conference (2026)
- **Reviewer**, Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)
- **Reviewer**, Paladyn: Journal of Behavioral Robotics
- **Reviewer**, Signal, Image, and Video Processing (Springer)
- **Reviewer**, International Conference on Advances in Computing, Communication, and Engineering

## Honors & Awards

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- Chancellor’s Graduate Research Travel Award, University of Nebraska–Lincoln (2026)
- Dean’s Fellowship, University of Nebraska–Lincoln (2025)
- Milton E. Mohr Fellowship (2024)
- David Swanson Memorial Travel Award (2023)
- Graduate Research Assistantship (2023)
- MHRD Fellowship (2014–2016)

## Technical Skills

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**Programming:** Python, PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn

**Deep Learning:** LLM fine-tuning (QLoRA, LoRA, PEFT), Transformers, CNNs, RNNs, Time-Series Models

**Tools:** Linux, Git, Docker, Slurm HPC, W&B, Optuna, LaTeX

**Domains:** NLP, Computer Vision, Behavioral Modeling, Multimodal Learning